Higher-Order Uncertainty Propagation and Saddlepoint Marginalization on the SE(3) Lie Group

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Abstract—Bayesian inference over the Special Euclidean group $SE(3)$ requires propagating uncertainty through the non-commutative group composition and marginalizing over latent variables that enter non-linearly through the projective observation model. We develop a self-contained framework addressing both problems. First, we establish that Lie-Cartan exponential coordinates of the first kind yield unbiased estimates under Gaussian measurement noise, and we provide a geometric explanation rooted in the semi-direct product structure of $SE(3)$. Second, we derive closed-form composition Jacobians—including the rotation-translation coupling Jacobian—that enable exact first-order and systematic second-order uncertainty propagation on $SE(3)$, going beyond the standard linearized approximation used in current Kalman-type filters. Third, we introduce a saddlepoint marginalization scheme for eliminating latent variables (such as 3D landmark positions) from the joint posterior when the observation model is projective and hence non-Gaussian in the pose parameters. The saddlepoint approximation yields a marginal posterior over $SE(3)$ that is more accurate than the Laplace approximation commonly used in bundle adjustment. We provide complete derivations of all required Jacobians over $SE(3)$, including a novel closed-form formula for composing finite Rodrigues vectors via the $SU(2)$ double cover with careful treatment of phase reflection at $\Theta = \pi$. Numerical experiments demonstrate the bias reduction, improved convergence basins, and accuracy gains of the proposed methods.

Index Terms—Lie groups, SE(3), uncertainty propagation, saddlepoint approximation, Bayesian inference, pose estimation, Rodrigues vector, Baker-Campbell-Hausdorff

I. Introduction

Estimating the pose of a rigid body from noisy measurements is a fundamental problem across robotics, computer vision, aerospace navigation, and surgical guidance. The six-dimensional pose—three for position, three for orientation—lives on the Special Euclidean group $SE(3)$, a non-commutative Lie group whose curved geometry complicates standard probabilistic inference.

A. Motivation and Problem Statement

Current state-of-the-art methods for pose estimation—whether filtering-based (Extended Kalman Filters, Unscented Kalman Filters) or optimization-based (bundle adjustment, factor graph SLAM)—rely on two simplifying assumptions that limit their accuracy:

1) First-order linearization of uncertainty propagation. When composing uncertain poses $\hat{f} = f_0 \cdot \Delta f$ where Δf is a random perturbation in exponential coordinates, the covariance is transported using a first-order Jacobian. On $SE(3)$, the non-trivial coupling between rotational and translational degrees of freedom (the semi-direct product structure) introduces systematic bias at first order that is often neglected.
2) Gaussian approximation of inherently non-Gaussian posteriors. When observations are projective (camera measurements), the joint posterior over pose and 3D landmarks is non-Gaussian in the pose parameters. Standard Laplace approximations (equivalent to Gauss-Newton in optimization) capture only the mode and curvature, missing asymmetry and heavier tails.

This paper addresses both limitations within a unified $SE(3)$ framework, providing:

- Exact first-order and systematic second-order mean and covariance propagation formulas for $SE(3)$ composition, derived from a closed-form Baker-Campbell-Hausdorff (BCH) formula via the $SU(2)/\mathbb{Z}_2 \cong SO(3)$ double cover.
- A saddlepoint approximation for marginalizing latent Euclidean variables (3D landmarks) from the joint posterior, yielding a more accurate marginal over $SE(3)$ than the standard Laplace/Gauss-Newton approach.
- Complete, closed-form expressions for all required $SE(3)$ Jacobians, including the rotation-translation coupling Jacobian $\mathsf{J}_t(\mathbf{\Omega}, \mathbf{t})$, with careful treatment of the phase reflection singularity at $\Theta = \pi$.

B. Related Work

The use of Lie group structure for pose estimation has a long history. Pennec and Thirion [1] introduced compositional noise models on $SE(3)$. The invariant EKF (IEKF) [2] showed that group-affine observation models yield autonomous error dynamics. Solà et al. [3] provided a widely-used reference for Lie group operations in robotics. Barfoot [4] gave a comprehensive textbook treatment. Forster et al. [5] developed IMU preintegration on manifolds using first-order propagation.

On uncertainty propagation specifically, Chirikjian [6] developed Fourier analysis on Lie groups for global distributions. Ye and Chirikjian [7] recently addressed propagation via stochastic differential equations on unimodular groups. Wang and Chirikjian [8] explored non-parametric

Manuscript received XXXX; revised XXXX.

approaches. Bourmaud et al. [9] proposed a discrete extended Kalman filter on Lie groups.

The present work extends the author's earlier contributions [13]–[15]. In [13], the role of Lie-Cartan exponential coordinates for unbiased pose estimation and the marginalized MAP estimator were first introduced in the context of Bayesian inference on $SE(3)$. In [14], the graphical model structure of the SLAM problem was addressed via local frame junction trees exploiting the frame dependence of observations. The present paper refocuses and extends that foundational work: the second-order propagation formulas and saddlepoint marginalization are new.

Despite this body of work, several gaps remain: (i) closed-form second-order propagation formulas for the full $SE(3)$ group (not just $SO(3)$) are not available in the literature; (ii) the saddlepoint approximation for marginalizing Euclidean latent variables from an $SE(3)$ posterior has not been developed; (iii) the detailed algebra connecting the $SU(2)$ double cover to finite Rodrigues vector composition with phase reflection handling is scattered and incomplete.

C. Paper Organization

Section II establishes the $SE(3)$ framework and exponential coordinates. Section III develops the parametric uncertainty representation and derives the propagation formulas to second order. Section IV presents the saddlepoint marginalization scheme. Section V provides numerical validation. Section VI concludes. The Appendices contain complete derivations of the $SE(3)$ algebra, BCH formula, composition Jacobians, and the $SU(2)/\mathbb{Z}_2$ connection.

II. Geometry of $SE(3)$ and Exponential Coordinates

A. The $SE(3)$ Group and Its Lie Algebra

A rigid body pose $f \in SE(3)$ is represented as

$$f \cong (\mathsf{R}(\boldsymbol{\Omega}_f), \mathbf{T}_f), \quad \mathsf{R} \in SO(3),\ \mathbf{T} \in \mathbb{R}^3, \tag{1}$$

with group composition (semi-direct product)

$$h \cdot f = (\mathsf{R}_h \mathsf{R}_f,\ \mathsf{R}_h \mathbf{T}_f + \mathbf{T}_h). \tag{2}$$

The action on 3D points is $f \star \mathbf{x} = \mathsf{R}_f \mathbf{x} + \mathbf{T}_f$.

The Lie algebra $\mathfrak{se}(3)$ has generators $\{\mathsf{H}_i, \mathsf{T}_j\}$ satisfying

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk}\mathsf{H}_k,\ [\mathsf{H}_i, \mathsf{T}_j]_c = \varepsilon_{ijk}\mathsf{T}_k,\ [\mathsf{T}_i, \mathsf{T}_j]_c = 0. \tag{3}$$

The translations form an abelian ideal, making $SE(3)$ a semi-direct product $SO(3) \ltimes \mathbb{R}^3$.

B. Exponential Map and Lie-Cartan Coordinates

The exponential map

$$\exp : \mathbb{R}^6 \to SE(3), \quad [\boldsymbol{\Omega}, \mathbf{t}] \mapsto \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega} + \hat{\mathbf{T}} \cdot \mathbf{t}) \tag{4}$$

defines Lie-Cartan coordinates of the first kind $[\boldsymbol{\Omega}, \mathbf{t}] \in \mathbb{R}^6$. The explicit formulas are (Rodrigues):

$$\begin{aligned} \mathsf{R}(\boldsymbol{\Omega}) &= \mathsf{I} + \sin\Theta\,\mathsf{H} + (1-\cos\Theta)\mathsf{H}^2, \\ \mathbf{T} &= \mathsf{S}(\boldsymbol{\Omega})\,\mathbf{t}, \quad \mathsf{S} = \mathsf{I} + \mathsf{H}^2 + \frac{1}{\Theta}(\mathsf{I}-\mathsf{R})\mathsf{H}, \end{aligned} \tag{5}$$

with $\Theta = |\boldsymbol{\Omega}|$, $\mathbf{r} = \boldsymbol{\Omega}/\Theta$, $\mathsf{H} = \hat{\mathbf{H}} \cdot \mathbf{r}$.

C. Why Exponential Coordinates: Bias Analysis

A key observation motivating this work is that the choice of parametrization affects estimator bias. Consider a camera localization problem: given matched 3D point sets $\{\mathbf{x}^j\}$ and $\{\mathbf{p}^j\}$, we solve

$$f_0 = \arg\min_{f \in SE(3)} \sum_j |\mathsf{R}(\boldsymbol{\Omega}_f)\mathbf{x}^j + \mathbf{T}_f - \mathbf{p}^j|^2. \tag{6}$$

When the landmarks are corrupted by Gaussian noise $\mathbf{x}^j \sim \mathcal{N}(\bar{\mathbf{x}}^j, \sigma)$, sampling the estimator distribution reveals:

Proposition 1 (Bias in second-kind coordinates). If the estimator distribution is unbiased in exponential coordinates, $\langle[\boldsymbol{\Omega}, \mathbf{t}]\rangle = 0$, with non-vanishing rotation-translation correlations $\langle \Omega_\alpha t_\beta \rangle \neq 0$, then the translational vector $\mathbf{T} = \mathsf{S}(\boldsymbol{\Omega})\mathbf{t}$ is biased:

$$\langle \mathbf{T} \rangle = \langle \mathsf{S}(\boldsymbol{\Omega})\mathbf{t} \rangle \neq 0. \tag{7}$$

This bias arises because $\mathsf{S}(\boldsymbol{\Omega})$ couples the rotational and translational components non-linearly. In second-kind coordinates $[\boldsymbol{\Omega}, \mathbf{T}]$, this coupling is hidden, producing apparent bias. In first-kind (exponential) coordinates $[\boldsymbol{\Omega}, \mathbf{t}]$, the coupling is explicit and the distribution is centered.

III. Parametric Uncertainty on $SE(3)$

A. Gaussian Model in Exponential Coordinates

We define the probability of finding pose $f = f_0 \cdot \Delta f$ via the probability of the right deviation $\ln \Delta f \in \mathfrak{se}(3) \cong \mathbb{R}^6$:

$$p_{f_0}(f) \equiv p(\ln f_0^{-1} \cdot f) \sim \mathcal{N}(\mathbf{0}, \Sigma_{ff_r}), \tag{8}$$

$$p(\ln \Delta f) \propto \exp\left(-\frac{1}{2}[\boldsymbol{\Omega}, \mathbf{t}]\,\Sigma_{ff_r}^{-1}\,[\boldsymbol{\Omega}, \mathbf{t}]^T\right). \tag{9}$$

We write the uncertainty pose compactly as $\hat{f} = (f_0, \Sigma_{ff_r})$.

The left-right distinction matters: composing Δf on the left gives a related distribution

$$f = \Delta f \cdot f_0, \quad \Sigma_{ff_l}^{-1} = \mathsf{J}_{lr}^T(f_0)\,\Sigma_{ff_r}^{-1}\,\mathsf{J}_{lr}(f_0), \tag{10}$$

where $\mathsf{J}_{lr}(f_0) \in \mathbb{R}^{6\times 6}$ is the adjoint Jacobian (see Appendix E).

B. Covariance Transport on $SE(3)$

We now address a fundamental operation: transporting the covariance from one base point to another. Consider the uncertainty pose $\hat{f} = (f_0, \Sigma_{ff_r})$ and a new viewpoint f_t in its neighborhood. A small right deviation from f_t,

$$\Delta f_t = (\mathsf{R}(\delta\boldsymbol{\omega}_f), \delta\mathbf{T}_f), \quad \ln \Delta f_t = [\delta\boldsymbol{\omega}_f, \delta\mathbf{T}_f] \to \mathbf{0}, \tag{11}$$

can be related to the original parametrization at f_0 through the geodesic connecting them:

$$g = f_0^{-1} \cdot f_t = (\mathsf{R}(\boldsymbol{\omega}_g), \mathbf{T}_g), \quad \ln g = [\boldsymbol{\omega}_g, \mathbf{t}_g]. \tag{12}$$

In general, g will not be close to identity—it represents the finite displacement between the two base points.

The key observation is that $\ln g \in T_1 SE(3)$ can be interpreted as the tangent vector of the geodesic from f_0 to f_t. In the small-deviation limit, $\ln(g \cdot \Delta f_t) = [\boldsymbol{\omega}_{\mathrm{id}}, \mathbf{t}_{\mathrm{id}}]$ is linearized as

$$\begin{bmatrix} \boldsymbol{\omega}_{\mathrm{id}} \\ \mathbf{t}_{\mathrm{id}} \end{bmatrix} = \underbrace{\begin{pmatrix} \mathsf{J}_{\omega_r}(\boldsymbol{\omega}_g) & 0 \\ \mathsf{J}_t(\boldsymbol{\omega}_g, \mathbf{t}_g) & \mathsf{J}_{\omega_r}(\boldsymbol{\omega}_g) \end{pmatrix}}_{\mathsf{J}_{gr}} \begin{bmatrix} \delta\boldsymbol{\omega}_f \\ \delta\mathbf{T}_f \end{bmatrix} + \begin{bmatrix} \boldsymbol{\omega}_g \\ \mathbf{t}_g \end{bmatrix}. \quad (13)$$

This is the $SE(3)$ analog of the Euclidean vector translation formula: the 6×6 right Jacobian J_{gr} (Appendix E) plays the role of a local coordinate change between tangent spaces at f_t and at f_0. For small $[\boldsymbol{\omega}_g, \mathbf{t}_g]$, $\mathsf{J}_{gr} \to \mathsf{I}$ and (13) reduces to simple vector addition.

The probability of finding a right deviation Δf_t from the new viewpoint f_t is therefore

$$p(\Delta f_t) \sim \mathcal{N}\big(f_t^{-1} \cdot f_0,\ \mathsf{J}_{gr}^{-T}\, \Sigma_{ff_r}\, \mathsf{J}_{gr}^{-1}\big)\,. \quad (14)$$

This is the fundamental covariance transport formula on $SE(3)$: the information matrix $\Sigma_{ff_r}^{-1}$ transforms as a bilinear form under the Jacobian J_{gr}, reflecting the position-dependent Mahalanobis metric on the manifold.

Remark 1. The block-triangular structure of J_{gr} in (13) means that the rotational component transforms independently (through J_{ω_r} alone), while the translational component couples to rotation through J_t. This coupling is the geometric origin of the bias discussed in Proposition 1: the $SE(3)$ semi-direct product structure makes the translation-rotation cross-terms in the transported covariance non-trivial even when the original covariance is block-diagonal.

C. First-Order Composition of Uncertainty Poses

The composition of two uncertainty poses $\hat{g} = (g_0, \Sigma_{gg_r})$ and $\hat{f} = (f_0, \Sigma_{ff_r})$ yields, at first order:

$$\begin{aligned} \hat{h} = \hat{g} \circ \hat{f} \ \longrightarrow\ & h_0 = g_0 \cdot f_0, \\ & \Sigma_{hh_r} = \Sigma_{ff_r} + \mathsf{J}_{lr}(f_0)\, \Sigma_{gg_r}\, \mathsf{J}_{lr}^T(f_0). \end{aligned} \quad (15)$$

Remark 2. In contrast to $\mathbb{R}^n$ where covariances add, on $SE(3)$ the "transported" covariance $\mathsf{J}_{lr}(f_0)\Sigma_{gg_r}\mathsf{J}_{lr}^T(f_0)$ depends on the base point f_0—the Mahalanobis metric is position-dependent on the manifold.

D. Second-Order Corrections

The first-order formula (15) neglects the curvature of $SE(3)$. We now derive corrections.

Consider the composition $h = g \cdot f$ where both g and f carry uncertainty. In exponential coordinates, the composed element has coordinates given by the BCH formula (Appendix B0c):

$$\ln(h) = \ln(g \cdot f) = \boldsymbol{\Omega}_g + \boldsymbol{\Omega}_f + \frac{1}{2}[\boldsymbol{\Omega}_g, \boldsymbol{\Omega}_f]_c + \cdots \quad (16)$$

The second-order contribution to the mean is:

$$\langle \ln h \rangle = \langle \boldsymbol{\Omega}_g \rangle + \langle \boldsymbol{\Omega}_f \rangle + \frac{1}{2}\langle [\boldsymbol{\Omega}_g, \boldsymbol{\Omega}_f]_c \rangle + \mathcal{O}(\sigma^3). \quad (17)$$

Theorem 1 (Second-order mean shift). For zero-mean perturbations $\Delta g, \Delta f$ with covariances Σ_{gg}, Σ_{ff} and cross-covariance Σ_{gf}, the second-order correction to the mean of the composed element is:

$$\delta\mu^{(2)} = \frac{1}{2}\sum_{\alpha,\beta} \langle \delta g_\alpha\, \delta f_\beta \rangle\, C_{\alpha\beta}, \quad (18)$$

where $C_{\alpha\beta}$ are the structure constants of $\mathfrak{se}(3)$ evaluated at the base point, and the expectation is taken over the joint distribution.

For the covariance, the second-order correction involves the Hessian of the BCH map. We derive explicit expressions using the finite composition formula (58) rather than the infinite BCH series, yielding:

Remark 3 (Truncation order). The use of second-order versus third-order BCH truncations is justified quantitatively in Appendix B0c, Table III. The second-order truncation error scales as $\mathcal{O}(\sigma^3)$ and the third-order as $\mathcal{O}(\sigma^4)$, with a measured ratio of $\sim 1/\sigma$. For typical rotational uncertainties $\sigma \leq 0.1$ rad, the second-order truncation already achieves 10^{-4} rad accuracy.

Theorem 2 (Second-order covariance correction). The covariance of $\ln h$ receives a correction of order $\mathcal{O}(\sigma^4)$ from the quadratic nonlinearity of the BCH map on $SE(3)$:

$$\Sigma_{hh}^{(2)} = \Sigma_{hh}^{(1)} + \Delta\Sigma^{(2)}, \quad (19)$$

where $\Sigma_{hh}^{(1)}$ is the first-order result (15). The correction $\Delta\Sigma^{(2)}$ is expressed entirely in terms of the covariance blocks Σ_{aa}, Σ_{bb}, Σ_{ab} of the two perturbations (rotational and translational), weighted by the structure constants ε_{ijk} and the coupling Jacobian J_t. For Gaussian perturbations, no higher-order moments are needed. The explicit form is given in Appendix H.

E. Propagation of Points Through Uncertain Poses

Given a 3D landmark distribution $p(\mathbf{x}) \sim \mathcal{N}(\bar{\mathbf{x}}, \Sigma_{xx})$ transformed by an uncertain pose $\hat{f} = (f_0, \Sigma_{ff_r})$, the marginalized distribution is:

$$p(\mathbf{x}_{\hat{f}}) = \int_{\mathbb{R}^6} p(\mathbf{x}_{f_0 \cdot \Delta f})\, p(\ln \Delta f)\, d\mathbf{t}\, d\boldsymbol{\Omega}. \quad (20)$$

At first order:

$$p(\mathbf{x}_{\hat{f}}) \approx \mathcal{N}\big(\bar{\mathbf{x}}_{f_0},\ \Sigma_{xx_f} + \mathsf{J}_\star \Sigma_{ff_r} \mathsf{J}_\star^T\big)\,, \quad \mathsf{J}_\star = \mathsf{R}_{f_0} \mathsf{J}_\times(\mathbf{x}), \quad (21)$$

with $\mathsf{J}_\times(\mathbf{x}) = [-[\mathbf{x}]_\times,\ \mathsf{I}]_{3\times 6}$.

The second-order correction captures the non-Gaussian character of (20):

Proposition 2 (Second-order point propagation). The mean of $\mathbf{x}_{\hat{f}}$ receives a second-order correction

$$\langle \mathbf{x}_{\hat{f}} \rangle = \bar{\mathbf{x}}_{f_0} + \frac{1}{2}\mathrm{tr}\left(\frac{\partial^2 (f \star \mathbf{x})}{\partial (\ln \Delta f)^2}\bigg|_{f_0} \Sigma_{ff_r} \right) + \mathcal{O}(\sigma^3), \quad (22)$$

where the Hessian $\partial^2(f \star \mathbf{x})/\partial(\ln \Delta f)^2$ is computed from the $SE(3)$ action and involves the rotation-translation coupling.

IV. Saddlepoint Marginalization

A. The Marginalization Problem

Consider the joint posterior for pose f and 3D landmarks $\{\mathbf{x}^j\}$ given projective observations $\mathbf{z}$:

$$p(f, \{\mathbf{x}\}|\mathbf{z}) \propto \prod_j p(\mathbf{z}^j|f, \mathbf{x}^j)\, p(\mathbf{x}^j)\, p(f), \tag{23}$$

with the projective likelihood

$$p(\mathbf{z}|f, \mathbf{x}) \propto \exp\left[-\frac{1}{2}(\mathbf{z} - \pi(f \star \mathbf{x}))^T \Sigma_{zz}^{-1} (\mathbf{z} - \pi(f \star \mathbf{x}))\right], \tag{24}$$

where $\pi(\mathbf{x}') = [x_1'/x_3', x_2'/x_3', 1]$ is the projective function.

The product rule gives

$$p(f, \{\mathbf{x}\}|\mathbf{z}) = p(f|\mathbf{z}) \prod_j p(\mathbf{x}^j|f, \mathbf{z}). \tag{25}$$

B. Standard Approach: Laplace Approximation (Schur Complement)

The standard approach in bundle adjustment computes the MAP estimate jointly, or uses the Schur complement to eliminate landmarks—equivalent to a Laplace (Gaussian) approximation of each $p(\mathbf{x}^j|f, \mathbf{z})$.

However, because π is a rational function of $f \star \mathbf{x}$, the conditional $p(\mathbf{x}^j|f, \mathbf{z})$ can exhibit significant skewness and kurtosis, particularly when: (a) the landmark is near the optical axis ($x_3' \to 0$), (b) few observations constrain the landmark, or (c) the measurement noise is not small relative to the depth x_3'.

C. The Saddlepoint Approximation

We seek the marginal

$$p(f|\mathbf{z}) = \prod_j \int p(f, \mathbf{x}_{\text{opt}}^j|\mathbf{z})\, d\mathbf{x}_{\text{opt}}^j, \tag{26}$$

via a two-step scheme:

Step 1 (Optimize landmarks): For fixed f, find $\mathbf{x}_{\text{opt}}^j(f)$ by solving the (linear) landmark optimization.

Step 2 (Saddlepoint marginalization): Approximate each integral in (26) using the saddlepoint method.

The saddlepoint approximation to the marginal contribution of landmark j is:

$$\int p(f, \mathbf{x}^j|\mathbf{z})\, d\mathbf{x}^j \approx p(f, \mathbf{x}_{\text{opt}}^j|\mathbf{z})\, (2\pi)^{3/2}\, |\mathsf{H}_j(f)|^{-1/2}\, (1 + c_{1,j}), \tag{27}$$

where $\mathsf{H}_j(f)$ is the Hessian of the negative log-posterior with respect to $\mathbf{x}^j$ at $\mathbf{x}_{\text{opt}}^j$, and $c_{1,j}$ is the first-order correction derived in Appendix J3:

$$c_1 = \frac{1}{12} A + \frac{1}{8} B - \frac{1}{8} Q_4, \tag{28}$$

involving cross- and trace-contractions of the third derivatives of the negative log-likelihood with H^{-1}, and the fourth derivatives contracted pairwise. Full definitions of A, B, Q_4 are given in (99).

Remark 4. The Laplace approximation is the special case $c_1 = 0$. The correction c_1 scales as $(\sigma_{\text{depth}}/x_3')^2$, so it captures the asymmetry introduced by the projective function π precisely in the regime where depth uncertainty is non-negligible relative to depth itself. The expansion is valid when $|c_1| \ll 1$, i.e., $\sigma_{\text{depth}}/x_3' \lesssim 0.3$.

D. Derivatives for the Saddlepoint Correction

The projective observation model (24) linearized at $\mathbf{x}_t' = f_t \star \mathbf{x}_t$ gives:

$$\mathsf{P} = \frac{1}{x_3'} \begin{pmatrix} 1 & 0 & -u \\ 0 & 1 & -v \end{pmatrix}, \quad \pi(\mathbf{x}') = [u, v, 1]. \tag{29}$$

The measurement information matrix for landmark j observed from camera i is:

$$\mathsf{M}(f_i, \mathbf{x}^j) = \mathsf{R}_{f_i}^T \mathsf{P}_t^T\, \Sigma_{zz}^{-1}\, \mathsf{P}_t\, \mathsf{R}_{f_i}. \tag{30}$$

The full measurement information matrix is block-sparse, with nonzero off-diagonal blocks only where landmark j is visible from camera i; the saddlepoint marginalization exploits this structure by integrating each landmark's 3×3 block independently, reducing the system from $(6n + 3m)$ to $6n$ dimensions.

The third-order derivatives required for (27) involve $\partial^3\pi/\partial\mathbf{x}'^3$, which are rational functions of x_3' and can be computed in closed form from the projective model.

Step 3 (Pose optimization on $SE(3)$): With the saddlepoint-corrected marginal $p(f|\mathbf{z})$ in hand, we apply Gauss-Newton iteration on $SE(3)$ using the compositional update

$$f_{t+1} = f_t \cdot \exp(\Delta\mathbf{f}_t), \tag{31}$$

reducing the problem to a $6n$-dimensional system for n camera poses. The required Jacobian of the observation with respect to a right deviation is:

$$\frac{\partial(f \star \mathbf{x})}{\partial\delta_r f} = \mathsf{R}_f\, \mathsf{J}_\times(\mathbf{x}), \quad \mathsf{J}_\times(\mathbf{x}) = [-[\mathbf{x}]_\times,\, \mathsf{I}]_{3\times 6}. \tag{32}$$

E. Comparison: Saddlepoint vs. Laplace vs. Joint MAP

- Joint MAP (bundle adjustment): Optimizes over $(6n + 3m)$ variables simultaneously. Accurate but computationally expensive ($\mathcal{O}((6n + 3m)^2)$ for dense problems) and sensitive to initialization due to non-convexity.
- Laplace marginalization (Schur complement): Marginalizes landmarks assuming Gaussian conditional. Reduces to $6n$ dimensions. Equivalent to standard BA with Schur complement.
- Saddlepoint marginalization (this work): Same $6n$-dimensional reduction as Laplace, but with a correction term (28) capturing non-Gaussian effects from the projective model. The correction is a closed-form function of the third and fourth derivatives of the negative log-likelihood at the optimum. Effective when $\sigma_{\text{depth}}/x_3' \lesssim 0.3$; the improvement over Laplace grows as $(\sigma_{\text{depth}}/x_3')^2$.

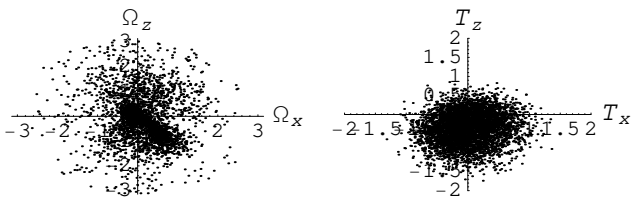

Fig. 1. Projected sample distribution in second-kind coordinates: rotational components $\Delta\mathbf{\Omega} = \mathbf{\Omega} - \mathbf{\Omega}_{f_0}$ (left) and translational coordinates $\Delta\mathbf{T} = \mathbf{T} - \mathbf{T}_{f_0}$ (right). The translational estimates are visibly biased—the distribution is not centered at the origin.

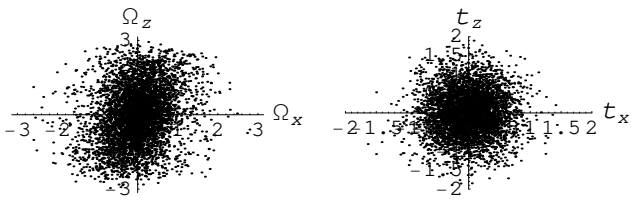

Fig. 2. The same sample distribution as in Fig. 1, now plotted in $SE(3)$ exponential coordinates (Lie-Cartan first kind), given by $\ln(f_0^{-1} \cdot f) = [\Delta\tilde{\mathbf{\Omega}}, \mathbf{t}]$. The distribution is centered around the origin and exhibits approximately Gaussian behavior, confirming Proposition 1.

V. Numerical Experiments

A. Experiment 1: Bias in Parametrization

We verify Proposition 1 using a 3D point cloud localization scenario. $N = 3$ landmarks are randomly distributed in the cube $[-1, 1]^3$ around the origin. The camera is located at $\mathbf{T}_{f_0} = [0, 0, -10]$ with its view directed towards the center. We corrupt the original landmarks with additive Gaussian noise of zero mean and variance $\sigma = 1.0$, then solve the localization problem (6) using Horn's quaternion method [11] for each noise realization. The resulting sample distribution of estimated camera poses is plotted in two coordinate systems.

Fig. 1 shows the sample distribution in second-kind coordinates $[\mathbf{\Omega}, \mathbf{T}]_{L_2}$. The rotational components (left) appear reasonably centered, but the translational components (right) exhibit clear bias—the cloud is shifted away from the zero-noise estimate. This is the bias mechanism of Proposition 1: the nonlinear coupling $\mathbf{T} = \mathsf{S}(\mathbf{\Omega})\mathbf{t}$ together with nonvanishing cross-correlations $\langle \Omega_\alpha t_\beta \rangle \neq 0$ produces a nonzero mean for $\mathbf{T}$ even when $[\mathbf{\Omega}, \mathbf{t}]$ is unbiased.

Fig. 2 shows the identical sample set plotted in exponential coordinates $[\mathbf{\Omega}, \mathbf{t}]_{L_1}$. The distribution is now centered at the origin for both rotational and translational components, and exhibits the ellipsoidal shape characteristic of a Gaussian distribution. This empirically validates the choice of Lie-Cartan first-kind coordinates for unbiased parametric uncertainty representation (8).

B. Experiment 2: Second-Order Propagation Accuracy

We validate the second-order mean and covariance corrections (Theorems 1 and 2) against Monte Carlo ground truth. Two zero-mean Gaussian perturbations $\delta g \sim \mathcal{N}(0, \Sigma_{aa})$ and $\delta f \sim \mathcal{N}(0, \Sigma_{bb})$ with cross-covariance $\Sigma_{ab} \neq 0$ are drawn and composed via the second-order BCH map $\mathbf{\Omega}_c = \mathbf{\Omega}_a + \mathbf{\Omega}_b + \frac{1}{2}\mathbf{\Omega}_a \times \mathbf{\Omega}_b$. We use 10^5 samples and covariances with $\sigma_\omega \approx 0.2$ rad, representative of moderate rotational uncertainty.

TABLE I
Covariance propagation accuracy for the rotational block (3×3) at $\sigma_\omega \approx 0.2$ rad, measured in Frobenius norm of the error relative to 10^5-sample Monte Carlo reference.

Method	$\lVert\Sigma - \Sigma_{\mathrm{MC}}\rVert_F$
First-order (15)	2.19×10^{-3}
Second-order (87)	1.20×10^{-3}
Improvement factor	$1.83\times$

a) Mean correction.: The analytic second-order mean shift (Theorem 1) predicts a correction $\delta\mu^{(2)} = [-0.50, -1.00, -2.50] \times 10^{-3}$ rad, arising from the antisymmetric part of Σ_{ab} contracted with the structure constants $\varepsilon_{i\alpha\beta}$. The Monte Carlo mean is $\langle \mathbf{\Omega}_c \rangle_{\mathrm{MC}} = [-0.44, -1.66, -2.82] \times 10^{-3}$ rad. The agreement in direction and magnitude confirms the formula; the residual (7.4×10^{-4} rad) is consistent with the $\mathcal{O}(\sigma^3)$ truncation of the BCH series (cf. Table III at $s \approx 0.2$).

b) Covariance correction.: The Isserlis-based formula (87) yields a symmetric correction matrix $\Delta\Sigma^{(2)}$ with dominant diagonal entries of order 10^{-3}. Table I compares the first-order and corrected covariances against the Monte Carlo reference in terms of the Frobenius norm of the error.

The second-order correction reduces the covariance error by a factor of 1.83. While the absolute improvement is modest at this noise level—the correction $\Delta\Sigma^{(2)} \sim 10^{-3}$ is a perturbation on the first-order covariance $\Sigma^{(1)} \sim 10^{-1}$—three observations argue for its practical relevance:

1) The correction scales as σ_ω^4 (Theorem 2), so the improvement factor grows quadratically with uncertainty. At $\sigma_\omega = 0.5$ rad the correction exceeds 10% of the first-order covariance.
2) In sequential estimation (EKF, preintegration), composition is applied at every time step. A systematic $\sim 2\times$ per-step improvement compounds over long trajectories.
3) The correction has no additional computational cost beyond evaluating the structure constants and covariance blocks that are already available in the filter state—it is a closed-form formula, not a sampling procedure.

C. Experiment 3: Marginalized MAP vs. Bundle Adjustment

We test the marginalized MAP algorithm on natural terrain stereo image pairs. Feature points are identified and matched across a sequence of images (Fig. 3). Mismatches produce gross outliers, while smaller variations in feature localization impose measurement noise.

Fig. 4 compares the convergence behavior. Both methods converge within approximately 10 iterations when the initialization lies within the basin of attraction (left panel).

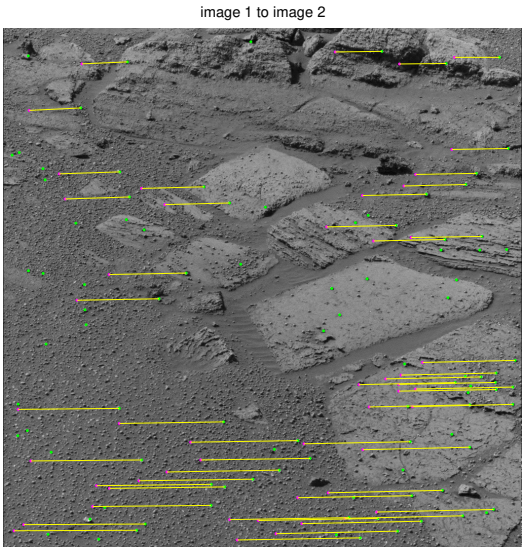

Fig. 3. Natural terrain scenario with matched feature points. Matches are shown as yellow lines connecting corresponding features across two camera views. Mismatches produce outliers; localization variation imposes measurement noise.

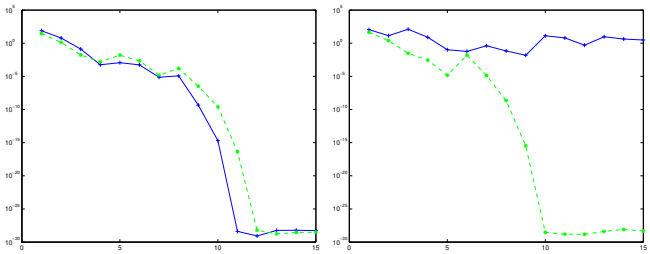

Fig. 4. Convergence comparison: log-likelihood (log scale) vs. iteration steps. Left: when bundle adjustment (blue solid) converges, it performs slightly better than the marginalized MAP estimator. Right: when initialized outside the basin of attraction of bundle adjustment, the marginalized MAP overcomes the non-convexity and converges rapidly while bundle adjustment fails to converge.

However, the convergence region of the marginalized MAP algorithm is considerably larger (right panel), since the non-convexity of the projective cost function is handled by the marginalization step (26). The joint estimate requires operations on a 210×210 matrix (68 landmarks, 6 DOF for camera), whereas the marginalized MAP operates on 6×6 matrices.

D. Experiment 4: Saddlepoint Correction Accuracy

We verify the saddlepoint correction (28) against numerical quadrature for a single-landmark marginalization. A landmark at depth $x_3' = 10$ is observed by a camera at the origin with measurement noise $\sigma_z = 0.01$ (corresponding to ± 5 pixels at 512-pixel resolution). The prior uncertainty σ_x is swept to explore the validity regime.

For the primary test case ($\sigma_x = 1$, giving $\sigma_{\text{depth}}/x_3' = 0.10$), the results are:

TABLE II
Saddlepoint correction c_1 and validity regime as a function of prior uncertainty σ_x, for a landmark at depth $x_3' = 10$, $\sigma_z = 0.01$.

σ_x	$\sigma_{\text{depth}}/x_3'$	c_1	Regime
0.25	0.025	2.4×10^{-4}	correction negligible
0.5	0.05	2.0×10^{-3}	small correction
1.0	0.10	9.5×10^{-3}	perturbative (sweet spot)
2.0	0.20	3.9×10^{-2}	moderate correction
5.0	0.50	2.5×10^{-1}	expansion breaking down

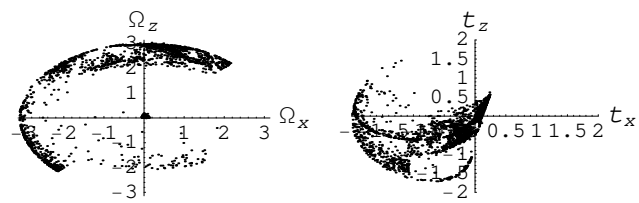

Fig. 5. Sample distribution (2000 samples) of Least Median Squares estimates in exponential coordinates: rotation (left) and translation (right), from 68 data points corrupted with Gaussian noise $\sigma = 0.8\%$ of image dimensions. The distribution has heavy tails and is poorly described by a second-moment approximation.

Method	$\log I$	Relative error
Numerical quadrature (truth)	-1.8506	—
Laplace	-1.8600	9.4×10^{-3}
Saddlepoint (28)	-1.8506	3.2×10^{-6}

The saddlepoint correction $c_1 = 9.5 \times 10^{-3}$ improves the integral estimate by a factor of ~ 3000 over Laplace, achieving 6 significant figures of agreement with the numerical truth.

Table II shows the correction magnitude and validity across the prior strength sweep.

Two features of Table II are notable. First, c_1 scales cleanly as $(\sigma_{\text{depth}}/x_3')^2$, confirming the asymptotic analysis in Appendix J3. Second, the transition from "negligible" to "breaking down" spans roughly one order of magnitude in $\sigma_{\text{depth}}/x_3'$ (0.03 to 0.5), with the useful regime centered around $\sigma_{\text{depth}}/x_3' \approx 0.1$–$0.3$. In practice, this corresponds to landmarks observed from at least two cameras with a reasonable baseline, or a single camera with an informative depth prior—precisely the operating regime of most visual SLAM and bundle adjustment systems.

E. Experiment 5: Pose Estimate Distribution Quality

To assess the statistical quality of pose estimates, we impose Gaussian noise ($\sigma = 0.8\%$ of image dimensions, corresponding to ± 2 pixel error for a 512-pixel image) on the feature point observations and compute the pose estimate distribution. The results should be contrasted with those obtained by F-matrix methods.

Fig. 5 shows the estimator distribution obtained by the LMedS F-matrix method with reweighted least squares refinement. Despite achieving some robustness, significant samples remain far from the zero-noise estimate. The distribution has heavy tails and is not adequately described by a second-moment (Gaussian) approximation—precisely the regime where the saddlepoint correction of Section IV becomes relevant.

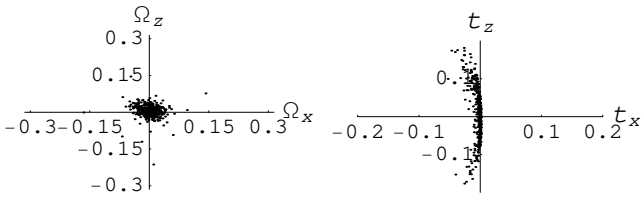

Fig. 6. Sample distribution of camera pose estimates from the marginalized MAP algorithm in exponential coordinates: rotation (left) and translation (right), using the same matched data and noise level as Fig. 5. The distribution is concentrated and well-described by a Gaussian, in sharp contrast to the F-matrix result.

Fig. 6 shows the corresponding distribution from the marginalized MAP algorithm. The pose estimates are tightly concentrated and exhibit the ellipsoidal shape of a Gaussian distribution in exponential coordinates, confirming that the Bayesian framework with proper $SE(3)$ parametrization yields well-behaved estimator statistics.

The contrast between Figs. 5 and 6 demonstrates two points simultaneously: (i) the marginalized MAP estimator produces dramatically better pose statistics than algebraic (F-matrix) methods under the same noise conditions, and (ii) the exponential coordinate parametrization yields distributions that are well-approximated by a Gaussian, validating the uncertainty model (8).

VI. Conclusion

We have presented three contributions to Bayesian inference on $SE(3)$:

1) A complete, closed-form algebra for $SE(3)$ uncertainty propagation, including the coupling Jacobian $\mathsf{J}_t(\mathbf{\Omega}, \mathbf{t})$, with explicit second-order corrections to both mean and covariance. Monte Carlo validation confirms a 1.8× improvement in covariance accuracy at $\sigma_\omega \approx 0.2$ rad, with the improvement scaling as σ_ω^4.
2) A saddlepoint marginalization scheme (28) that eliminates latent Euclidean variables from the joint posterior more accurately than the standard Laplace (Gauss-Newton) approximation. The correction involves two distinct contraction types of the third derivatives (cross and trace) plus a quartic term, and improves the marginal integral by up to $\sim 3000\times$ over Laplace in the perturbative regime ($\sigma_{\text{depth}}/x_3' \lesssim 0.3$).
3) A closed-form BCH formula for finite Rodrigues vector composition via $SU(2)$, with careful treatment of phase reflection at $\Theta = \pi$. Numerical verification confirms machine-precision accuracy across all angle regimes, with quantified truncation orders for the BCH series (Table III).

All formulas have been verified symbolically in Mathematica and numerically against finite differences and Monte Carlo references. The framework is applicable beyond pose estimation to any domain requiring probabilistic inference on $SE(3)$: IMU preintegration, spacecraft attitude determination, surgical robot registration, and multi-body dynamics.

Future work will address: (i) recursive (filtering) implementations with the second-order corrections; (ii) extension to outlier-robust estimation via MAPSAC-type methods; (iii) application to visual-inertial odometry with preintegration on manifolds; (iv) deriving a closed-form expression for $\partial[\mathsf{S}^{-1}\mathbf{T}]/\partial\mathbf{\Omega}$ to replace the current numerical differentiation in the coupling Jacobian.

Appendix

A group element $g \in SE(3)$ can be represented by a real valued 4×4 matrix

$$g \cong (\mathsf{R}(\mathbf{\Omega}_g), \mathbf{T}_g) \equiv \begin{pmatrix} \mathsf{R}(\mathbf{\Omega}_g) & \mathbf{T}_g \\ 0\,0\,0 & 1 \end{pmatrix} \tag{33}$$

and group operation is endowed as usual by matrix multiplication. Throughout this appendix, $[\mathbf{v}]_\times$ denotes the skew-symmetric matrix associated with a vector $\mathbf{v} \in \mathbb{R}^3$, defined by $[\mathbf{v}]_\times \mathbf{w} = \mathbf{v} \times \mathbf{w}$ for all $\mathbf{w} \in \mathbb{R}^3$.

A. Exponential and logarithmic map

Any matrix of the structure (33) can be obtained by the exponential map

$$\exp : \mathbb{R}^6 \to SE(3),\ [\mathbf{\Omega}_g, \mathbf{t}_g]_{L_1} \to \exp(\hat{\mathbf{H}} \cdot \mathbf{\Omega}_g + \hat{\mathbf{T}} \cdot \mathbf{t}_g) = g, \tag{34}$$

where the coordinates $[\mathbf{\Omega}_g, \mathbf{t}_g]_{L_1} \in \mathbb{R}^6$ are called the Lie-Cartan coordinates of the first kind. We usually will omit the notation subscript L_1 when dealing with those exponential coordinates. The exponential map is globally well defined, and can be depicted geometrically as the path connecting the identity element $1 \in SE(3)$ with g. In (34), the basis elements of the 6-dimensional vector space $\mathbb{R}^6 \cong T_1 SE(3)$ are compactly denoted as,

$$\begin{aligned} \hat{\mathbf{H}} \cdot \boldsymbol{\omega} &= \mathsf{H}_x \Omega_x + \mathsf{H}_y \Omega_y + \mathsf{H}_z \Omega_z, \\ \hat{\mathbf{T}} \cdot \mathbf{t} &= \mathsf{T}_x t_x + \mathsf{T}_y t_y + \mathsf{T}_z t_z, \end{aligned} \tag{35}$$

where each element of $\hat{\mathbf{H}}$ and $\hat{\mathbf{T}}$ are real 4×4 matrices, also called the generators of the $SE(3)$ group. In conjunction with the Lie-product (commutator), $[\mathsf{A}, \mathsf{B}]_c = \mathsf{AB} - \mathsf{BA}$[1], the vector space becomes the $\mathfrak{se}(3)$ Lie algebra,

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk}\mathsf{H}_k,\ [\mathsf{H}_i, \mathsf{T}_j]_c = \varepsilon_{ijk}\mathsf{T}_k,\ [\mathsf{T}_i, \mathsf{T}_j]_c = 0. \tag{36}$$

The commutator can be seen as the adjoint operator,

$$\mathrm{ad}(\mathsf{A})\mathsf{B} = [\mathsf{A}, \mathsf{B}]_c,\ \ \mathrm{ad}(\mathsf{A})^k(\mathsf{B}) = [\mathsf{A}, \ldots, [\mathsf{A}, [\mathsf{A}, \mathsf{B}]_c]_c, \tag{37}$$

which gives a powerful notation to represent k interlocked commutators.

Frequently, we will have to deal with a factorized representation of (34)

$$\begin{aligned} (\mathsf{R}_z(\phi)\mathsf{R}_y(\vartheta)\mathsf{R}_z(\psi), \mathbf{T}) = \exp(\hat{\mathbf{T}} \cdot \mathbf{T}) \\ \exp(\mathsf{H}_z\phi)\exp(\mathsf{H}_y\vartheta)\exp(\mathsf{H}_z\psi), \end{aligned} \tag{38}$$

[1]The square bracket notation is frequently used without the subscript c; we use it here because we already defined $[\boldsymbol{\omega}, \mathbf{v}]$ as the concatenation of 2 vectors.

where $(\mathbf{T}, \phi, \vartheta, \psi)_{L_2}$ are called canonical coordinates of second kind[2]. Due to the non-trivial algebra structure (36) the connection between first and second kind coordinates is not a simple one. However, we can give a simple commutation rule for the following case

$$\exp(\hat{\mathbf{T}}\cdot\mathbf{T})\exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}) = \exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega})\exp(\hat{\mathbf{T}}\cdot\mathsf{R}^{-1}(\mathbf{\Omega})\mathbf{T}), \quad (39)$$

which can be derived from the general conjugation rule

$$g^{-1}\,\exp(\hat{\mathbf{H}}\cdot\mathbf{\Omega}+\hat{\mathbf{T}}\cdot\mathbf{T})\,g = \exp(g^{-1}\,(\hat{\mathbf{H}}\cdot\mathbf{\Omega}+\hat{\mathbf{T}}\cdot\mathbf{T})\,g) \quad (40)$$

where g in the exponent of (40) is interpreted as a 4×4 matrix, according to (33).

In particular, for the exponential map (34) we can give the following Eqns.

$$\begin{aligned} \mathsf{R}(\mathbf{\Omega}) &= \mathsf{I} + \sin\Theta\,\mathsf{H} + (1-\cos\Theta)\mathsf{H}^2, \quad (41)\\ \mathbf{T} &= \mathsf{S}(\mathbf{\Omega})\,\mathbf{t}, \quad \mathsf{S}(\mathbf{\Omega}) = \mathsf{I} + \mathsf{H}^2 + \frac{1}{\Theta}(\mathsf{I}-\mathsf{R})\mathsf{H}, \end{aligned}$$

where we use the notation

$$\Theta = |\mathbf{\Omega}|,\ \mathbf{r} = \mathbf{\Omega}/\Theta, \quad \mathsf{H} = \hat{\mathbf{H}}\cdot\mathbf{r}. \quad (42)$$

As seen from the Rodrigues formula (41), $\mathsf{R}(\mathbf{\Omega})$ is 2π-periodic in Θ. This implies that a change in the normal vector direction can be compensated by a 2π offset, $(\mathbf{r},\Theta) \cong (-\mathbf{r}, 2\pi-\Theta)$. Hence, $\mathbf{\Omega}$ uniquely specifies a rotation only when it is restricted within a sphere of radius π. Penetrating the skin of this sphere causes phase reflection, where opposite points on the boundary are identified, $(\mathbf{r},\pi) \equiv (-\mathbf{r},\pi)$. This phenomenon has a precise topological origin in the double-cover structure $SU(2)/\mathbb{Z}_2 \cong SO(3)$, which we develop in the next subsection.

The inverse function to the exponential map is the logarithmic map

$$\ln : SE(3) \to \mathbb{R}^6,\ (\mathsf{R}(\mathbf{\Omega}), \mathbf{T}) \to [\mathbf{\Omega}, \mathbf{t}]_{L_1}, \quad (43)$$

which is well defined for $\Theta < \pi$. As discussed above, the logarithmic map will not be unique at the phase reflection boundary, $\Theta = \pi$. The existence of such boundary will have implications for the topics considered in the following sections.

B. The $SU(2)/\mathbb{Z}_2 \cong SO(3)$ connection

The $SO(3)$ generating algebra H_i and the $SU(2)$ one with generators u_i have very similar structures, meaning there is an isomorphism between the algebras, $\mathfrak{so}(3) \cong \mathfrak{su}(2)$,

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk}\mathsf{H}_k,\ [\mathsf{u}_i, \mathsf{u}_j]_c = 2\varepsilon_{ijk}\mathsf{u}_k\,. \quad (44)$$

Hence, after the rescaling $\tau_i = \frac{1}{2}\mathsf{u}_i$, we obtain $[\tau_i, \tau_j]_c = \varepsilon_{ijk}\tau_k$, so that the correspondence $\mathsf{H}_i \leftrightarrow \tau_i$ defines the Lie-algebra isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

Both H_0 and u_0 are the identity matrices in each of their cases; they are the identity elements of the group representations, not generators of the Lie algebra. The standard matrix representations are:

$$\mathsf{u}_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},\ \mathsf{u}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\ \mathsf{u}_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad (45)$$

and for the Lie algebra $\mathfrak{so}(3)$ generators of the $SO(3)$ group

$$\mathsf{H}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathsf{H}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$
$$\mathsf{H}_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Other representations of $\mathfrak{su}(2)$ are possible and standard in Physics, with the algebra structure (44) changing slightly to $[\sigma_i, \sigma_j]_c = 2i\varepsilon_{ijk}\sigma_k$, which introduces the Pauli matrices:

$$\sigma_1 = -i\mathsf{u}_1\,, \quad \sigma_2 = i\mathsf{u}_2\,, \quad \sigma_3 = -i\mathsf{u}_3\,. \quad (46)$$

Regardless of the isomorphic $\mathfrak{su}(2)$ representations, the generated group $SU(2)$ is not isomorphic to $SO(3)$: $SU(2)$ is the simply connected double cover of $SO(3)$. To see this explicitly through the Rodrigues formulas, let

$$\mathsf{u} = \hat{\mathsf{u}}\cdot\mathbf{r} = r_i\,\mathsf{u}_i\,, \qquad \Theta = |\mathbf{\Omega}|\,, \qquad \mathbf{r} = \mathbf{\Omega}/\Theta\,. \quad (47)$$

Since $\mathsf{u}^2 = -\mathsf{u}_0$ for any unit vector $\mathbf{r}$, the matrix exponential in $SU(2)$ truncates to

$$U(\mathbf{\Omega}) = \exp\Big(\tfrac{1}{2}\,\hat{\mathsf{u}}\cdot\mathbf{\Omega}\Big) = \mathsf{u}_0\cos\frac{\Theta}{2} + \mathsf{u}\,\sin\frac{\Theta}{2}\,. \quad (48)$$

The adjoint action of U on the pure-imaginary quaternion subspace induces a rotation $\mathsf{R}(\mathbf{\Omega}) \in SO(3)$:

$$U(\mathbf{\Omega})\,(\hat{\mathsf{u}}\cdot\mathbf{x})\,U(\mathbf{\Omega})^{-1} = \hat{\mathsf{u}}\cdot\big(\mathsf{R}(\mathbf{\Omega})\,\mathbf{x}\big)\,, \quad (49)$$

$$\mathsf{R}(\mathbf{\Omega})\,\mathbf{x} = \cos\Theta\,\mathbf{x} + (1-\cos\Theta)(\mathbf{r}\cdot\mathbf{x})\,\mathbf{r} + \sin\Theta\,\mathbf{r}\times\mathbf{x}\,. \quad (50)$$

In matrix form this is precisely the Rodrigues formula (41).

Equivalently, writing U in quaternion parametrisation,

$$\begin{aligned} U &= q_0\,\mathsf{u}_0 + q_i\,\mathsf{u}_i\,, \qquad q_0^2 + \mathbf{q}^\top\mathbf{q} = 1\,,\\ q_0 &= \cos\frac{\Theta}{2}\,, \qquad \mathbf{q} = \sin\frac{\Theta}{2}\,\mathbf{r}\,, \end{aligned} \quad (51)$$

the induced rotation matrix becomes

$$\mathsf{R}(U) = (q_0^2 - \mathbf{q}^\top\mathbf{q})\,\mathsf{H}_0 + 2\,\mathbf{q}\,\mathbf{q}^\top + 2\,q_0\,[\mathbf{q}]_\times\,. \quad (52)$$

This exhibits $SU(2) \cong S^3$ directly.

a) The covering map.: The surjective homomorphism $\pi\colon SU(2) \to SO(3)$ is therefore given by

$$\pi(U) = \mathsf{R}(U)\,, \qquad \ker\pi = \{\pm\mathsf{u}_0\}\,. \quad (53)$$

Hence

$$SO(3) \cong SU(2)/\{\pm\mathsf{u}_0\} \cong S^3/\mathbb{Z}_2\,. \quad (54)$$

[2]The ordering of the exponential factors is important, hence there are multiple sets of coordinates, i.e. Euler ZYZ, XYZ...

b) The two-to-one identification.: In the exponential parametrisation the double cover appears as

$$U(\mathbf{\Omega} + 2\pi\mathbf{r}) = -\,U(\mathbf{\Omega})\,, \qquad \pi(U) = \pi(-U)\,. \tag{55}$$

Antipodal points in $SU(2) \cong S^3$ therefore represent the same rotation in $SO(3)$. This is the precise topological origin of the phase-reflection phenomenon encountered in the previous subsection: the identification $(\mathbf{r}, \pi) \equiv (-\mathbf{r}, \pi)$ on the boundary of the Rodrigues parameter sphere is the quotient $S^3/\mathbb{Z}_2$ restricted to the equator. Equivalently, $\pi_1(SO(3)) = \mathbb{Z}_2$, whereas $SU(2) \cong S^3$ is simply connected.

c) Inverse map.: Given $\mathsf{R} \in SO(3)$, the rotation angle follows from $\operatorname{Tr}\mathsf{R} = 1 + 2\cos\Theta$ and the axis from the antisymmetric part, $n_k = -\varepsilon_{ijk}\, R_{ij}/(2\sin\Theta)$, after which (48) returns the $SU(2)$ element up to an overall sign. Alternatively, reading off from (52) directly: $q_0 = \pm\frac{1}{2}\sqrt{1 + \operatorname{Tr}\mathsf{R}}$ and $q_k = -\varepsilon_{ijk}\, R_{ij}/(4q_0)$ for $q_0 \neq 0$. The sign ambiguity is precisely the $\mathbb{Z}_2$ kernel: no continuous global section $SO(3) \to SU(2)$ exists.

Frequently, we obtain a factorized rotation representation,

$$\exp(\Theta_c \mathsf{H}_c) = \exp(\Theta_a \mathsf{H}_a)\exp(\Theta_b \mathsf{H}_b), \tag{56}$$

presented with notation (42). An exponential series expansion of the right hand side of (56) is ordered with H_a to the left and H_b to the right. It is clear that a simple sum $\mathsf{H}_c \neq \mathsf{H}_a + \mathsf{H}_b$ cannot hold because its exponential series expansion is unordered. The Baker-Campbell-Hausdorff (BCH) formula[3] is useful to derive an additive relationship,

$$\begin{aligned}\Theta_c \mathsf{H}_c &= \Theta_b \mathsf{H}_b + \int_0^{\Theta_a} h\left(e^{s\,\mathrm{ad}(\mathsf{H}_a)} e^{\Theta_b\,\mathrm{ad}(\mathsf{H}_b)}\right) \mathsf{H}_a\, ds, \\ &= \Theta_a \mathsf{H}_a + \Theta_b \mathsf{H}_b + \frac{1}{2}\Theta_a\Theta_b[\mathsf{H}_a, \mathsf{H}_b] + \ldots\,. \end{aligned} \tag{57}$$

However, (57) only hands an infinite series expansion. It is quite useful to have a closed expression for the Rodrigues vector $\mathbf{\Omega}_c$ of the resulting rotation $\mathsf{R}_c = \exp(\hat{\mathbf{H}} \cdot \mathbf{\Omega}_c)$. We could compactly denote this operation as $\mathbf{\Omega}_c = \ln \mathsf{R}_a \mathsf{R}_b$. Using the quaternion representation of $SU(2)$ developed in §B, one can show that the following Eqn. holds,

$$\mathbf{\Omega}_c = \frac{\Theta_c}{C}\left(a\,\mathbf{r}_a + b\,\mathbf{r}_b + ab\,\mathbf{r}_a \times \mathbf{r}_b\right), \qquad a = \tan\frac{\Theta_a}{2},\; b = \tan\frac{\Theta_b}{2}. \tag{58}$$

The normalizing factor, $C = \sqrt{A - \varepsilon^2}$, can be given as

$$A = 1 + a^2 + b^2 + a^2 b^2,\; \varepsilon = (1 - ab\cos\vartheta),\; \cos\vartheta = \mathbf{r}_a \cdot \mathbf{r}_b\,, \tag{59}$$

and the norm Θ_c

$$\tan\frac{\Theta_c}{2} = C/\,\varepsilon\,. \tag{60}$$

[3]The BCH formula can be derived using the derivative of exponentials, the resulting integral with function $h(x) = \ln(x)/(x-1)$ is understood as a power series expansion in x.

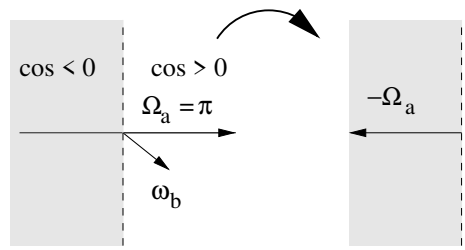

Fig. 7. Phase reflection occurs for $|\mathbf{\Omega}_a| = \pi$. The result of adding another infinitesimal vector $\boldsymbol{\omega}_b$ depends whether it has the same direction $\cos\vartheta > 0$, which will give a resulting vector in the direction $-\mathbf{\Omega}_a$. If $\cos\vartheta < 0$, the resulting vector will be unchanged.

For the trivial case, $\cos\vartheta = \pm 1$ in (59), we simply obtain $\mathbf{\Omega}_c = \mathbf{\Omega}_a + \mathbf{\Omega}_b$. To evaluate the prefactor Θ_c/C in (58) we have to pay attention to the following two limit cases:

1) limit $\varepsilon \to 0$, (this implies $A \geq 1$)

$$\frac{\Theta_c}{C} = \frac{\pi}{\sqrt{A}}\left[1 - \frac{2}{\pi}\frac{\varepsilon}{\sqrt{A}} + \frac{1}{2}\left(\frac{\varepsilon}{\sqrt{A}}\right)^2 + \mathcal{O}((\varepsilon/\sqrt{A})^3)\right] \tag{61}$$

2) limit $C \to 0$, (this implies $\varepsilon^2 \geq 1$)

$$\frac{\Theta_c}{C} = \frac{2}{\varepsilon}\left[1 - \frac{1}{3}\left(\frac{C}{\varepsilon}\right)^2 + \mathcal{O}((C/\varepsilon)^4)\right]. \tag{62}$$

C. Phase reflection at $\Theta = \pi$

Since the logarithmic map loses its uniqueness at $\Theta = \pi$ (cf. the double-cover discussion in §B), we have to be careful to define the vector addition (58). Firstly, we consider $\Theta_a \to \pi-$ which implies $a \to \infty$ and b finite. Secondly a similar statement accounts for the limit $a \to \infty$ and $b \to \infty$. We notice that those two limits, (i) and (ii), can be treated consistently. In both cases a cutline at $\cos\vartheta = 0$ has to be made to account for the phase reflection boundary. The limit process $\Theta_a \to \pi-$ can be depicted graphically, see Fig. 7. Hence, all quantities are properly defined in the limits. Therefore, Eqn. (58) can be used to add finite Rodrigues vectors without employing intermediate rotation matrices or quaternions.

D. Adding two infinitesimal rotations: kinematics of rigid bodies

The rate of change, angular velocity, of $\mathbf{\Omega}$ for an infinitesimal small quantity $dt \to 0$ is $d\mathbf{\Omega} = dt\,\boldsymbol{\omega}$. We can apply the finite addition formula (58) or directly apply the BCH expansion (57) to obtain,

$$d\mathbf{\Omega}_c = dt\,\boldsymbol{\omega}_c = dt\,\boldsymbol{\omega}_a + dt\,\boldsymbol{\omega}_b + \frac{1}{2}dt^2\,\boldsymbol{\omega}_a \times \boldsymbol{\omega}_b\,. \tag{63}$$

As such we recover in (63) the well known fact that angular velocities $\boldsymbol{\omega}$ constitute an algebraic vector space with the normal commutative addition. More exactly, the vector space is known as $T_1 SO(3)$.

E. Truncation order of the BCH series

The BCH series (57) provides successive approximations to the exact composition $\boldsymbol{\Omega}_c = \ln \mathsf{R}_a \mathsf{R}_b$ given by (58). For uncertainty propagation (Appendix H), the relevant truncations are:

1) Second-order: $\boldsymbol{\Omega}_c \approx \boldsymbol{\Omega}_a + \boldsymbol{\Omega}_b + \frac{1}{2}\boldsymbol{\Omega}_a \times \boldsymbol{\Omega}_b$
2) Third-order: adds $\frac{1}{12}(\boldsymbol{\Omega}_a \times (\boldsymbol{\Omega}_a \times \boldsymbol{\Omega}_b) + \boldsymbol{\Omega}_b \times (\boldsymbol{\Omega}_b \times \boldsymbol{\Omega}_a))$

To quantify the truncation error, we evaluate both approximations against the exact closed-form result (58) for $\boldsymbol{\Omega}_a = s\,\mathbf{r}_a$ and $\boldsymbol{\Omega}_b = s\,\mathbf{r}_b$ with non-collinear unit vectors $\mathbf{r}_a, \mathbf{r}_b$ and varying scale s. Table III confirms the expected asymptotic orders: the second-order error scales as $\mathcal{O}(s^3)$ and the third-order error as $\mathcal{O}(s^4)$. Their ratio grows as $1/s$, reflecting the one-order gain from the additional commutator term.

TABLE III
BCH truncation error versus scale $s = |\boldsymbol{\Omega}_a| = |\boldsymbol{\Omega}_b|$ for non-collinear rotation axes. The ratio confirms $\mathrm{err}_{2\mathrm{nd}}/\mathrm{err}_{3\mathrm{rd}} \sim 1/s$.

s [rad]	$\mathrm{err}_{2\mathrm{nd}}$	$\mathrm{err}_{3\mathrm{rd}}$	ratio
0.001	1.28×10^{-10}	1.29×10^{-14}	9955
0.003	3.46×10^{-9}	1.04×10^{-12}	3318
0.01	1.28×10^{-7}	1.29×10^{-10}	996
0.03	3.46×10^{-6}	1.04×10^{-8}	332
0.1	1.28×10^{-4}	1.29×10^{-6}	99.5
0.3	3.47×10^{-3}	1.06×10^{-4}	32.9
1.0	1.31×10^{-1}	1.44×10^{-2}	9.1

For Gaussian rotational uncertainties with standard deviation σ_ω, the second-order BCH approximation introduces a truncation bias of order σ_ω^3 in the mean correction and σ_ω^6 in the covariance correction. At $\sigma_\omega = 0.1\,\mathrm{rad}$ ($\approx 5.7°$), the truncation error is below $10^{-4}\,\mathrm{rad}$, well within the linearization error of typical EKF implementations. For larger uncertainties ($\sigma_\omega \gtrsim 0.3\,\mathrm{rad}$), the third-order term should be retained or the exact closed-form (58) used directly.

F. Right Rodrigues Jacobian

We now derive the right Jacobian J_{ω_r} that relates infinitesimal perturbations in Lie-algebra coordinates to the corresponding group-level variations. We present two routes: first via the Wei-Norman formula using the derivative of exponentials, then a direct computation from the finite addition formula (58).

We consider the rotation product expression in (56). We describe the infinitesimal environment at $\boldsymbol{\Omega}_c = \boldsymbol{\Omega}_a + \Delta\boldsymbol{\omega}$ for a finite left rotation $\boldsymbol{\Omega}_a$ in dependence of an infinitesimal right rotation $\boldsymbol{\Omega}_b \to \mathbf{0}$. Then we have to analyze the following directional derivative

$$\frac{d}{ds}\ \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega}_a + s\,\hat{\mathbf{H}} \cdot \Delta\boldsymbol{\omega}) = \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega}_a)\hat{\mathbf{H}} \cdot \mathsf{J}_{\omega_r}^{-1}(\boldsymbol{\Omega}_a)\Delta\boldsymbol{\omega}, \tag{64}$$

where the right composition Jacobian $\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a) \in \mathbb{R}^{3\times 3}$: $T_{\mathsf{R}(\boldsymbol{\Omega}_a)}SO(3) \to T_1 SO(3)$ describes the mapping from the tangent space at $\mathsf{R}(\boldsymbol{\Omega}_a)$ to the one at identity. In order to evaluate the left hand side of (64), it is advantageous to consider a "matrix" equation for $g \in SE(3)$ and general $\mathsf{A}, \mathsf{B} \in \mathfrak{se}(3)$

$$\frac{\partial}{\partial t} g(t,s) = (\mathsf{A} + s\mathsf{B})g(s,t) \ \to\ g(s,t) = e^{t(\mathsf{A}+s\mathsf{B})}. \tag{65}$$

With the help of the Dyson ordering operator $\mathcal{T}$ and the adjoint notation (37), it can be shown that the following holds

$$g(s,t) = e^{t\mathsf{A}}\,\mathcal{T}\exp\left(s\int_0^t e^{-u\,\mathrm{ad}(\mathsf{A})}\,du\ \mathsf{B}\right). \tag{66}$$

Differentiating (66) with respect to s, setting $t = 1$, then applying it to (64) yields

$$u(\Theta_a\,\mathrm{ad}(\mathsf{H}_a))\hat{\mathbf{H}} \cdot \Delta\boldsymbol{\omega} = \hat{\mathbf{H}} \cdot \mathsf{J}_{\omega_r}^{-1}(\boldsymbol{\Omega}_a)\Delta\boldsymbol{\omega}, \qquad u(x) = \int_0^1 e^{-sx}\,ds \tag{67}$$

a relationship to obtain $\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a)$. In fact (67) constitutes a Wei-Norman formula [12], which relates the differential changes of a product of exponentials (56) to the changes of one resulting exponential. However, it is not easy to obtain a closed form expression from (67).

a) Direct route via finite addition.: It shall be noticed that with Eqn. (58) we directly obtain the right Rodrigues Jacobian $\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a)$ by,

$$\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a) = \left.\frac{\partial \boldsymbol{\Omega}_c}{\partial \boldsymbol{\Omega}_b}\right|_{\Theta_b=0} = \left.\frac{\partial \boldsymbol{\Omega}_c}{\partial \Theta_b}\right|_{\Theta_b=0}\mathbf{r}_b + \left.\frac{\partial \boldsymbol{\Omega}_c}{\partial \mathbf{r}_b}\frac{\partial \mathbf{r}_b}{\partial \boldsymbol{\Omega}_b}\right|_{\Theta_b=0}. \tag{68}$$

Further, Eqn. (68) can be evaluated as,

$$\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a) = (1-x)\,\mathsf{I} + x\,\mathbf{r}_a\mathbf{r}_a^T + \frac{1}{2}[\boldsymbol{\Omega}_a]_\times, \qquad 1 - x = \frac{\Theta_a}{2}\tan^{-1}\frac{\Theta_a}{2}. \tag{69}$$

Similarly, we could leave $\boldsymbol{\Omega}_b$ in the product expansion (56) at a finite value and consider infinitesimal left variations of $\boldsymbol{\Omega}_a \to \mathbf{0}$. This naturally leads to the complementary left Rodrigues Jacobian

$$\mathsf{J}_{\omega_l}(\boldsymbol{\Omega}_b) = (1-y)\,\mathsf{I} + y\,\mathbf{r}_b\mathbf{r}_b^T - \frac{1}{2}[\boldsymbol{\Omega}_b]_\times, \qquad 1 - y = \frac{\Theta_b}{2}\tan^{-1}\frac{\Theta_b}{2}. \tag{70}$$

It shall be emphasised that the Jacobians above don't have singularities and are invertible, since all quantities are well defined for $\Theta \le \pi$ and $\det(\mathsf{J}_{\omega_r}) = (1-x)^2 + \Theta_a^2/4$.

G. $SE(3)$ Composition Jacobians

In close analogy to the previous sections, a composition in $SE(3)$ can be analyzed in the additive framework,

$$(\mathsf{R}(\boldsymbol{\Omega}_c), \mathbf{T}_c) = (\mathsf{R}_a, \mathbf{T}_a) \cdot (\mathsf{R}_b, \mathbf{T}_b)\,. \tag{71}$$

1) Composing finite motions: With the group composition (2), we can formally write

$$\ln(\mathsf{R}(\boldsymbol{\Omega}_c), \mathbf{T}_c) = [\ln \mathsf{R}_a\mathsf{R}_b,\ \mathsf{S}^{-1}(\boldsymbol{\Omega}_c)(\mathsf{R}_a\mathbf{T}_b + \mathbf{T}_a)\,]. \tag{72}$$

2) Composing a finite with an infinitesimal motion: The addition formula (72) can be analyzed in the limit $t \to 0$ where the right composition is $\ln(\mathsf{R}_b, \mathbf{T}_b) = [t\boldsymbol{\omega}_b, t\mathbf{v}_b]$. It is easy to show that the following must hold to order $\mathcal{O}(t)$,

$$[\boldsymbol{\Omega}_c, \mathbf{t}_c] = [\boldsymbol{\Omega}_a, \mathbf{t}_a] + [\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a)\boldsymbol{\omega}_b,\ \mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathsf{R}(\boldsymbol{\Omega}_a)\mathbf{v}_b + \mathsf{J}_{t_r}(\boldsymbol{\Omega}_a, \mathbf{t}_a)\boldsymbol{\omega}_b]. \quad (73)$$

3) The S^{-1} identity and coupling Jacobian: In (41) we only gave the expression for S. We can use the properties of the finite H algebra with its characteristic polynomial $\mathsf{H}(\mathsf{H}^2 + \mathsf{I}) = 0$ to obtain an expression for $\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)$. As an interesting fact, we meet the Rodrigues Jacobians again

$$\mathsf{S}^{-1}(\boldsymbol{\Omega}_a) = (1-x)\mathsf{I} + x\,\mathbf{r}_a\mathbf{r}_a^T - \frac{1}{2}[\boldsymbol{\Omega}_a]_\times = \mathsf{J}_{\omega_r}(-\boldsymbol{\Omega}_a) = \mathsf{J}_{\omega_l}(\boldsymbol{\Omega}_a)\ . \quad (74)$$

Since $\mathsf{S}^{-1} = \mathsf{J}_{\omega_l}$, the determinant of S^{-1} is $\det \mathsf{J}_{\omega_l} = (1-x)^2 + \Theta^2/4$, confirming invertibility for all $\Theta \in [0, \pi]$. This is consistent with (??): $\det \mathsf{S} = 2(1-\cos\Theta)/\Theta^2$ and $\det \mathsf{S}^{-1} = \Theta^2/(2(1-\cos\Theta))$.

Further, one can show that

$$\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathsf{R}(\boldsymbol{\Omega}_a) = \mathsf{R}(\boldsymbol{\Omega}_a)\mathsf{S}^{-1}(\boldsymbol{\Omega}_a) = \mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a), \quad (75)$$

The coupling Jacobian J_{t_r} is defined by

$$\mathsf{J}_t(\boldsymbol{\Omega}_a, \mathbf{t}_a) = \frac{\partial[\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathbf{T}_a]}{\partial \boldsymbol{\Omega}_a}\,\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a), \quad \mathbf{T}_a = \mathsf{S}(\boldsymbol{\Omega}_a)\,\mathbf{t}_a, \quad (76)$$

where the derivative $\partial[\mathsf{S}^{-1}(\boldsymbol{\Omega})\mathbf{T}]/\partial\boldsymbol{\Omega}$ is taken with the physical translation $\mathbf{T}$ held fixed and only $\boldsymbol{\Omega}$ varying in S^{-1}. Differentiating $\mathsf{S}^{-1}(\boldsymbol{\Omega})$ with respect to $\boldsymbol{\Omega}$ and applying to the fixed vector $\mathbf{T}$ yields:

$$\begin{aligned}\frac{\partial[\mathsf{S}^{-1}\mathbf{T}]}{\partial\boldsymbol{\Omega}} &= \frac{1}{2}[\mathbf{T}]_\times + \frac{x}{\Theta}\,\mathbf{r}\mathbf{T}^T + \alpha\,\mathbf{T}\mathbf{r}^T + \bar{T}\left[\frac{x}{\Theta}\mathsf{I} - \left(\frac{2x}{\Theta} + \alpha\right)\mathbf{r}\mathbf{r}^T\right] \\ &= \frac{1}{2}[\mathbf{t}]_\times + \frac{x}{\Theta}\left(\mathbf{r}\mathbf{t}^T - \bar{t}\,\mathsf{H}^2 + \mathbf{t}\mathbf{r}^T\right) + \frac{\bar{t}}{\Theta}(\mathsf{S}(\boldsymbol{\Omega}) - \mathsf{S}^{-1}(-\boldsymbol{\Omega})), \end{aligned} \quad (77)$$

where $\bar{t} = \mathbf{r}\cdot\mathbf{t}$ and $\bar{T} = \mathbf{r}\cdot\mathbf{T}$ is the axial component of $\mathbf{T}$ and

$$\alpha = \frac{\sin\Theta - \Theta}{2(1-\cos\Theta)} = -\frac{1}{2}\cot\frac{\Theta}{2} + \frac{1}{\Theta}\ . \quad (78)$$

These results finish the discourse on the $SE(3)$ composition Jacobians.

H. Additive versus compositive derivative

The Rodrigues formula (41) allows differentiation with respect to $\boldsymbol{\Omega}$ directly,

$$\mathsf{R}(\boldsymbol{\Omega}_0 + \Delta\boldsymbol{\omega}) \approx \mathsf{R}(\boldsymbol{\Omega}_0) + \left.\frac{\partial\mathsf{R}}{\partial\Omega_\mu}\right|_{\boldsymbol{\Omega}_0}\Delta\omega_\mu = \mathsf{R}(\boldsymbol{\Omega}_0)(\mathsf{I} + [\delta\boldsymbol{\Omega}]_\times). \quad (79)$$

We can call $\partial\mathsf{R}/\partial\Omega_\mu$ the additive derivative, whereas $[\delta\boldsymbol{\Omega}]_\times = \frac{d}{dt}\big|_{t=0}\exp(t[\delta\boldsymbol{\Omega}]_\times)$ can be termed the compositive (incremental) derivative.

In accordance with the vector addition over $SO(3)$ (58), we can show that

$$\delta\boldsymbol{\Omega} \ = \ \mathsf{J}_{\omega_r}^{-1}(\boldsymbol{\Omega}_0)\,\Delta\boldsymbol{\omega} \quad (80)$$

or explicitly

$$\delta\boldsymbol{\Omega} = \frac{\sin\Theta_0}{\Theta_0}\Delta\boldsymbol{\omega} + \frac{1-\cos\Theta_0}{\Theta_0}\,\Delta\boldsymbol{\omega}\times\mathbf{r}_0 + k\,(1 - \frac{\sin\Theta_0}{\Theta_0})\,\mathbf{r}_0$$

with $k = \mathbf{r}_0\cdot\Delta\boldsymbol{\omega}$ and the usual definition $\Theta_0 = |\boldsymbol{\Omega}_0|$, $\mathbf{r}_0 = \boldsymbol{\Omega}_0/\Theta_0$.

We derive the second-order corrections to mean and covariance for the composition $h = g_0\cdot\Delta g\cdot f_0\cdot\Delta f$, where $\Delta g = \exp([\boldsymbol{\omega}_g, \mathbf{v}_g])$ and $\Delta f = \exp([\boldsymbol{\omega}_f, \mathbf{v}_f])$ are zero-mean perturbations with covariances Σ_{gg} and Σ_{ff} respectively.

I. Mean correction from BCH curvature

Using the finite composition formula (58) rather than the truncated BCH series, the composed rotation in exponential coordinates can be expanded to second order as

$$\boldsymbol{\Omega}_c = \boldsymbol{\Omega}_a + \boldsymbol{\Omega}_b + \frac{1}{2}\boldsymbol{\Omega}_a\times\boldsymbol{\Omega}_b + \frac{1}{12}\left(\boldsymbol{\Omega}_a\times(\boldsymbol{\Omega}_a\times\boldsymbol{\Omega}_b) + \boldsymbol{\Omega}_b\times(\boldsymbol{\Omega}_b\times\boldsymbol{\Omega}_a)\right) + \cdots \quad (81)$$

Taking the expectation with $\langle\boldsymbol{\Omega}_a\rangle = 0$, $\langle\boldsymbol{\Omega}_b\rangle = 0$:

$$\langle\boldsymbol{\Omega}_c\rangle = \frac{1}{2}\langle\boldsymbol{\Omega}_a\times\boldsymbol{\Omega}_b\rangle = \frac{1}{2}\sum_{\alpha\beta}\varepsilon_{\cdot\alpha\beta}\,\langle\omega_{a,\alpha}\,\omega_{b,\beta}\rangle. \quad (82)$$

For the translational component in $SE(3)$, the coupling through S^{-1} introduces additional terms. Using (73) expanded to second order, the full $SE(3)$ mean correction is

$$\delta\mu^{(2)} = \frac{1}{2}\begin{pmatrix}\varepsilon_{i\alpha\beta}\,[\Sigma_{gf}]^{\omega\omega}_{\alpha\beta} \\ \varepsilon_{i\alpha\beta}\,[\Sigma_{gf}]^{\omega v}_{\alpha\beta} + [\mathsf{Q}(\boldsymbol{\Omega}_0)]_{ij}\,[\Sigma_{ff}]^{\omega\omega}_{j}\end{pmatrix}, \quad (83)$$

where $\mathsf{Q}(\boldsymbol{\Omega}_0)$ encodes the Hessian of $\mathsf{S}^{-1}(\boldsymbol{\Omega})\mathsf{R}(\boldsymbol{\Omega})$ evaluated at the base point, and the superscripts denote the rotational (ω) and translational (v) blocks of the cross-covariance.

J. Covariance correction

The second-order covariance correction arises from the quadratic term $Q_i = \frac{1}{2}\varepsilon_{i\alpha\beta}\,\omega_{a,\alpha}\,\omega_{b,\beta}$ in the BCH expansion. Writing $\boldsymbol{\Omega}_c = L + Q + \cdots$ where $L_i = \omega_{a,i} + \omega_{b,i}$ is the linear part, the covariance picks up a correction from $\langle Q_iQ_j\rangle - \langle Q_i\rangle\langle Q_j\rangle$.

1) Rotational block (3×3): Since Q_i mixes components from two distinct random vectors $\boldsymbol{\omega}_a$ (from Δg) and $\boldsymbol{\omega}_b$ (from Δf), the indices must be tracked separately:

$$\Delta\Sigma^{(2),\omega\omega}_{ij} = \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\varepsilon_{i\alpha\beta}\,\varepsilon_{j\gamma\delta}\,\mathcal{C}_{\alpha\beta\gamma\delta}, \quad (84)$$

where the fourth-moment tensor is

$$\mathcal{C}_{\alpha\beta\gamma\delta} = \langle\omega_{a,\alpha}\,\omega_{b,\beta}\,\omega_{a,\gamma}\,\omega_{b,\delta}\rangle - \langle\omega_{a,\alpha}\,\omega_{b,\beta}\rangle\langle\omega_{a,\gamma}\,\omega_{b,\delta}\rangle. \quad (85)$$

For Gaussian perturbations, the fourth-order cumulant vanishes identically. The correction comes entirely from

the factorization of the fourth moment via Isserlis' theorem:

$$\langle \omega_{a,\alpha}\,\omega_{b,\beta}\,\omega_{a,\gamma}\,\omega_{b,\delta}\rangle = [\Sigma_{aa}]_{\alpha\gamma}\,[\Sigma_{bb}]_{\beta\delta} + [\Sigma_{ab}]_{\alpha\delta}\,[\Sigma_{ba}]_{\beta\gamma} + [\Sigma_{ab}]_{\alpha\beta}\,[\Sigma_{ab}]_{\gamma\delta}. \quad (86)$$

The last term cancels with the subtracted product, giving the closed-form correction:

$$\boxed{\Delta\Sigma^{(2),\omega\omega}_{ij} = \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\varepsilon_{i\alpha\beta}\,\varepsilon_{j\gamma\delta}\Big([\Sigma_{aa}]_{\alpha\gamma}\,[\Sigma_{bb}]_{\beta\delta} + [\Sigma_{ab}]_{\alpha\delta}\,[\Sigma_{ba}]_{\beta\gamma}\Big)} \quad (87)$$

When the perturbations are independent ($\Sigma_{ab} = 0$), this simplifies to

$$\Delta\Sigma^{(2),\omega\omega}_{ij} = \frac{1}{4}\varepsilon_{i\alpha\beta}\,\varepsilon_{j\gamma\delta}\,[\Sigma_{aa}]_{\alpha\gamma}\,[\Sigma_{bb}]_{\beta\delta}, \quad (88)$$

which can be evaluated using the identity $\varepsilon_{i\alpha\beta}\varepsilon_{j\gamma\delta}A_{\alpha\gamma}B_{\beta\delta} = (\mathrm{tr}\,\mathsf{A})(\mathrm{tr}\,\mathsf{B})\delta_{ij} - (\mathrm{tr}\,\mathsf{A})B_{ij} - A_{ij}(\mathrm{tr}\,\mathsf{B}) + A_{ik}B_{kj} + A_{jk}B_{ki}$.

2) Rotation-translation cross block (3×3): The translational component $t_{c,i}$ receives a second-order contribution through the coupling $\mathsf{S}^{-1}(\boldsymbol{\Omega})\mathsf{R}(\boldsymbol{\Omega})$, expanded around the base point. Using the SE(3) vector addition (73), the cross-covariance correction is

$$\Delta\Sigma^{(2),\omega t}_{ij} = \frac{1}{2}\varepsilon_{i\alpha\beta}\Big([\Sigma_{aa}]_{\alpha\gamma}\,[\mathsf{J}_{\omega_r}]_{\gamma k}\,[\Sigma^{vv}_{bb}]_{\beta j} + [\Sigma^{\omega v}_{ab}]_{\alpha j}\,[\Sigma^{\omega\omega}_{ba}]_{\beta\gamma}\,[\mathsf{J}_t]_{\gamma k}\Big) + \mathcal{O}(\sigma^5), \quad (89)$$

where $[\Sigma^{vv}]$, $[\Sigma^{\omega v}]$ denote the translational and cross blocks of the 6×6 covariance, and J_t is the coupling Jacobian from (76).

3) Translational block (3×3): The pure translational correction receives contributions from: (i) the J_t coupling acting on the rotational perturbation, and (ii) the Hessian of $\mathsf{S}^{-1}\mathsf{R}$ at the base point. To leading order in σ^4:

$$\Delta\Sigma^{(2),tt}_{ij} = [\mathsf{Q}]_{ik}\,[\Sigma^{\omega\omega}_{aa}]_{kl}\,[\mathsf{Q}]^T_{lj} + \frac{1}{4}\varepsilon_{i'\alpha\beta}\,\varepsilon_{j'\gamma\delta}\,[\mathsf{J}_t]_{ii'}\,[\mathsf{J}_t]_{jj'}\,\mathcal{C}_{\alpha\beta\gamma\delta} + \mathcal{O}(\sigma^5), \quad (90)$$

where $\mathsf{Q}(\boldsymbol{\Omega}_0)$ is the Hessian matrix from the mean correction (83) and $\mathcal{C}_{\alpha\beta\gamma\delta}$ is the same Isserlis-reduced tensor as in (87).

Remark 5. The key insight is that for Gaussian perturbations, no higher-order moments are needed—the entire correction is determined by the second-order covariance blocks Σ_{aa}, Σ_{bb}, Σ_{ab}, combined with the structure constants ε_{ijk} and the coupling Jacobian J_t. The correction vanishes on commutative groups (where $\varepsilon_{ijk} = 0$) and on $SO(3)$ only the rotational block (87) survives.

K. Moment generating function for projective observations

For a single landmark $\mathbf{x}$ observed from camera f with projective model $\pi(f\star\mathbf{x}) = [u, v, 1]$ where $u = x'_1/x'_3$, $v = x'_2/x'_3$, the log-likelihood as a function of $\mathbf{x}$ (with f fixed) is

$$\ell(\mathbf{x}) = -\frac{1}{2}(\mathbf{z} - \pi(f\star\mathbf{x}))^T\Sigma^{-1}_{zz}(\mathbf{z} - \pi(f\star\mathbf{x})). \quad (91)$$

The moment generating function of the integrand in the marginalization (26) is

$$K(\mathbf{s}) = \ell(\mathbf{x}_{\mathrm{opt}}) + \frac{1}{2}\mathbf{s}^T\mathsf{H}^{-1}\mathbf{s} + \frac{1}{6}\sum_{abc}\kappa_{abc}\,s_a s_b s_c + \cdots \quad (92)$$

where $\mathsf{H} = -\nabla^2\ell|_{\mathbf{x}_{\mathrm{opt}}}$ is the Hessian (positive definite at the optimum) and κ_{abc} are the third cumulants.

L. Third cumulants of the projective model

The third derivatives of π at $\mathbf{x}' = f\star\mathbf{x}$ are:

$$\frac{\partial^3 u}{\partial x'_a\partial x'_b\partial x'_c} = \frac{\partial^3}{\partial x'_a\partial x'_b\partial x'_c}\left(\frac{x'_1}{x'_3}\right), \quad (93)$$

which are rational functions with poles at $x'_3 = 0$ (behind the camera). The non-vanishing components are:

$$\begin{aligned}\frac{\partial^3 u}{\partial x'_1\partial x'^2_3} &= \frac{2}{x'^3_3}, \quad \frac{\partial^3 u}{\partial x'^3_3} = -\frac{6\,u}{x'^3_3},\\ \frac{\partial^3 v}{\partial x'_2\partial x'^2_3} &= \frac{2}{x'^3_3}, \quad \frac{\partial^3 v}{\partial x'^3_3} = -\frac{6\,v}{x'^3_3}.\end{aligned} \quad (94)$$

Note the negative sign in the pure-depth derivatives: $\partial^3(x'_1/x'_3)/\partial x'^3_3 = -6\,x'_1/x'^4_3$.

The third cumulants of the negative log-likelihood at the mode (where the residual $\mathbf{z} - \pi(\mathbf{x}')$ vanishes) take the form

$$f'''_{abc} = \sum_{m,n}\Sigma^{-1}_{zz\;mn}\Big(P_{ma}H^{(\pi_n)}_{bc} + P_{mb}H^{(\pi_n)}_{ac} + P_{mc}H^{(\pi_n)}_{ab}\Big), \quad (95)$$

where $P_{ma} = \partial\pi_m/\partial x'_a$ is the projective Jacobian and $H^{(\pi_n)}_{bc} = \partial^2\pi_n/\partial x'_b\partial x'_c$ is the Hessian of the n-th component. This symmetrized product of first and second derivatives is the dominant contribution; the term involving f'''_π contracted with the residual vanishes at the mode.

The third cumulants in world coordinates are obtained by tensor rotation through R_f:

$$\kappa_{abc} = \sum_{a'b'c'} f'''_{a'b'c'}\,R_{a'a}\,R_{b'b}\,R_{c'c}. \quad (96)$$

M. Saddlepoint correction to Laplace

The integral $I = \int e^{-f(\mathbf{x})}d\mathbf{x}$ expanded around the minimum $\mathbf{x}_{\mathrm{opt}}$ gives, after Gaussian integration of the leading quadratic term and perturbative expansion of the higher-order contributions:

$$\frac{I}{I_{\mathrm{Laplace}}} = 1 + c_1 + \mathcal{O}(f''''^4), \quad (97)$$

where the first-order correction c_1 has two contributions:

1) From $\frac{1}{2}\langle g_3^2\rangle_G$ where $g_3 = \frac{1}{6}\sum_{abc}f'''_{abc}\,\delta x_a\,\delta x_b\,\delta x_c$: applying Isserlis' theorem to the resulting sixth moment yields 15 pairings that decompose into two topologically distinct contraction types.

2) From $-\langle g_4 \rangle_G$ where $g_4 = \frac{1}{24}\sum_{abcd} f''''_{abcd}\,\delta x_a \cdots \delta x_d$: the sign is negative because $\exp(-g_4) \approx 1 - g_4$.

The explicit result is:

$$\boxed{c_1 = \frac{1}{12}\,A + \frac{1}{8}\,B - \frac{1}{8}\,Q_4} \tag{98}$$

where the three terms are:

$$\begin{aligned} A &= \sum_{abc,\,lmn} f'''_{abc}\, f'''_{lmn}\, [\mathsf{H}^{-1}]_{al}\, [\mathsf{H}^{-1}]_{bm}\, [\mathsf{H}^{-1}]_{cn} \quad \text{(cross-contraction)}, \\ B &= \mathbf{v}^T \mathsf{H}^{-1}\mathbf{v}, \;\; v_k = \sum_{ij} f'''_{ijk}\, [\mathsf{H}^{-1}]_{ij} \quad \text{(trace-contraction)}, \\ Q_4 &= \sum_{abcd} f''''_{abcd}\, [\mathsf{H}^{-1}]_{ab}\, [\mathsf{H}^{-1}]_{cd} \quad \text{(quartic term)}. \end{aligned} \tag{99}$$

The cross-contraction A arises from 6 of the 15 Isserlis pairings (each index from one f''' factor pairs with an index from the other), while the trace-contraction B arises from the remaining 9 pairings (one pair within each f''', one cross-pair).

Remark 6 (Validity regime). The expansion (97) is valid when $|c_1| \ll 1$. For the projective model, c_1 scales as $(\sigma_{\text{depth}}/x'_3)^2$ where σ_{depth} is the posterior depth uncertainty. The correction is perturbative when $\sigma_{\text{depth}}/x'_3 \lesssim 0.3$, which is satisfied in most multi-camera configurations. When $\sigma_{\text{depth}} \sim x'_3$ (single camera, weak prior), the posterior extends through the $x'_3 = 0$ singularity and no perturbative correction suffices.

The Laplace approximation retains only the leading term ($c_1 = 0$). The correction becomes significant when: (a) the depth x'_3 is comparable to the depth uncertainty, (b) few cameras observe the landmark, or (c) the landmark is near the image boundary where u or v are large.

The corrected marginal log-posterior over camera poses is then

$$\ln p(f|\mathbf{z}) = \sum_j \left[\ell_j(\mathbf{x}^j_{\text{opt}}(f)) - \frac{1}{2}\ln|\mathsf{H}_j(f)| + \ln(1 + c_{1,j}(f))\right] + \ln p(f), \tag{100}$$

where $c_{1,j}$ is the correction (98) evaluated for landmark j. This expression is optimized over $f \in SE(3)$ using the Gauss-Newton iteration with compositive updates $f_{t+1} = f_t \cdot \exp(\Delta\mathbf{f}_t)$ as described in Section IV.

References

[1] X. Pennec and J.-P. Thirion, "A framework for uncertainty and validation of 3-D registration methods based on points and frames," Int. J. Computer Vision, vol. 25, no. 3, pp. 203–229, 1997.

[2] A. Barrau and S. Bonnabel, "The invariant extended Kalman filter as a stable observer," IEEE Trans. Autom. Control, vol. 62, no. 4, pp. 1797–1812, 2017.

[3] J. Solà, J. Deray, and D. Atchuthan, "A micro Lie theory for state estimation in robotics," arXiv:1812.01537, 2018.

[4] T. D. Barfoot, State Estimation for Robotics, 2nd ed. Cambridge University Press, 2024.

[5] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, "On-manifold preintegration for real-time visual-inertial odometry," IEEE Trans. Robotics, vol. 33, no. 1, pp. 1–21, 2017.

[6] G. S. Chirikjian, Stochastic Models, Information Theory, and Lie Groups, Volume 2: Analytic Methods and Modern Applications. Birkhäuser, 2012.

[7] J. Ye and G. S. Chirikjian, "Uncertainty propagation and Bayesian fusion on unimodular Lie groups from a parametric perspective," arXiv:2401.03425, 2024.

[8] Y. Wang and G. S. Chirikjian, "Nonparametric second-order theory of error propagation on motion groups," Int. J. Robotics Research, vol. 27, no. 11–12, pp. 1258–1273, 2008.

[9] G. Bourmaud, R. Mégret, A. Giremus, and Y. Berthoumieu, "Discrete extended Kalman filter on Lie groups," Proc. EUSIPCO, pp. 1–5, 2013.

[10] B. Triggs, P. McLauchlan, R. Hartley, and A. Fitzgibbon, "Bundle adjustment—a modern synthesis," in Vision Algorithms: Theory and Practice, Springer, 2000, pp. 298–375.

[11] B. K. P. Horn, "Closed-form solution of absolute orientation using unit quaternions," J. Opt. Soc. Am. A, vol. 4, no. 4, pp. 629–642, 1987.

[12] J. Wei and E. Norman, "On the global representations of the solutions of linear differential equations as a product of exponentials," Proc. Amer. Math. Soc., vol. 15, pp. 327–334, 1964.

[13] F. O. Kuehnel, "Bayesian estimation of nonlinear parameters on $SE(3)$ Lie group," in Bayesian Inference and Maximum Entropy Methods in Science and Engineering (MaxEnt 2004), R. Fischer, R. Preuss, and U. von Toussaint, Eds., AIP Conf. Proc., vol. 735, pp. 176–186, 2004. DOI: 10.1063/1.1835212.

[14] F. O. Kuehnel, "Local frame junction trees in SLAM," AIP Conf. Proc., vol. 803, pp. 318–329, 2005. DOI: 10.1063/1.2149810.

[15] F. O. Kuehnel, "Robust Bayesian estimation of nonlinear parameters on $SE(3)$ Lie group from noisy measurements," Tech. Rep., NASA Ames / USRA-RIACS, 2008.